Project Design Phase

Problem – Solution Fit Template

Date	15 February 2026
Team ID	LTVIP2026TMIDS73502
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	2 Marks

Problem – Solution Fit Template:

People in flood-prone areas lack an easy and quick way to predict severe flood risk based on rainfall data. Most existing forecasts are general and do not provide probability-based prediction tailored to specific inputs.

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.

Template:

1. CUSTOMER SEGMENT(S) CS

- Residents in flood-prone areas
- Farmers
- Local government authorities
- Disaster management teams
- Environmental researchers

2. JOBS-TO-BE-DONE / J&P

- Check flood risk before heavy rainfall
- Make early safety decisions
- Reduce property and crop loss.
- Get quick flood probability estimation

5. AVAILABLE SOLUTIONS AS

- Government weather forecasts
- News reports
- Manual rainfall tracking

Limitation:

- Not predictive
- Not personalized
- No probability-bassed risk analysis

2. JOBS-TO-BE-DONE / PROBLEMS J&P

- Check flood risk before heavy rainfall
- Make early safety decisions
- Reduce property and crop loss
- Get quick flood probability estimation

3. TRIGGERS TR

- Continuous heavy rainfall
- Monsoon season alerts
- News about nearby floods
- Get quick flood probability levels

6. CUSTOMER CONSTRAINTS CC

- Limited technical knowledge
- Lack of real-time analytical tools
- Dependence on general forecasts
- Limited internet access in rural areas

Limitation:

- Not predictive
- Not personalized
- No probability-based risk analysis

3. TRIGGERS TR

- Continuous heavy rainfall
- Monsoon season alerts
- News about nearby floods
- Sudden increase in rainfal levels

4. PROBLEM ROOT CAUSE RC

- High rainfall variability
- Climate change effects
- Lack of accessible predictive systems
- Delayed information from traditiona

8. BEHAVIOUR BE

- Users check data during rainy season
- Users search flood alerts online
- Users make decisions based on news reports
- Users may share predictions with family/community

4 EMOTIONS: BEFORE / AFTER EM

- Before:
- Fear
- Uncertainty
- Confusion

- AFTER:
- Awareness
- Preparedness
- Confidence in decision-making

5. YOUR SOLUTION SL

- Web-based Flood Prediction System
- Uses Machine Learning (XGBoost)
- Provides severe flood probability instantly
- Simple and easy-to-use interface
- Can be extended with real-time weather data

8. CHANNELS of BEHAVIOUR CH

- Web browser (mobile/desktop)
- Local deployment for academic use
- Future: integration with government portals

10. YOUR SOLUTION SL

- Web-based Flood Prediction System
- Uses Machine Learning (XGBoost)
- Provides severe flood probability instanty